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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies <u>25</u>	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution <u>25</u>	Correct implementation with minor issues <u>25</u>	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration <u>20</u>	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results <u>10</u>	Good analysis with reasonable interpretation <u>10</u>	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results <u>9</u>	Organized documentation with necessary details <u>9</u>	Basic documentation with missing details	Poor or incomplete documentation

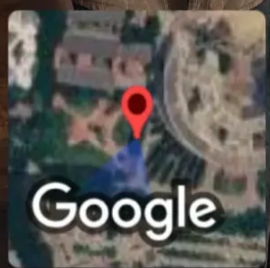
\$ 89 / 100



Satellite Internet Communication using the OSI Model CSA-0720

Presented by:
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GPS Map Camera



Mevalurkuppam, Tamil Nadu,
India 

22g8+c47, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.025983° Long 80.015693°

Thursday, 06/08/2026 10:39 AM GMT +05:30



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(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Satellite Internet Communication using the OSI Model

Presented by:Prajit.M.V(192521088)

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Introduction



- Internet service provided through communication satellites.
- Connects remote and rural areas.
- Uses satellites orbiting the Earth.
- Provides global internet connectivity.
 - Examples: Starlink, OneWeb, HughesNet

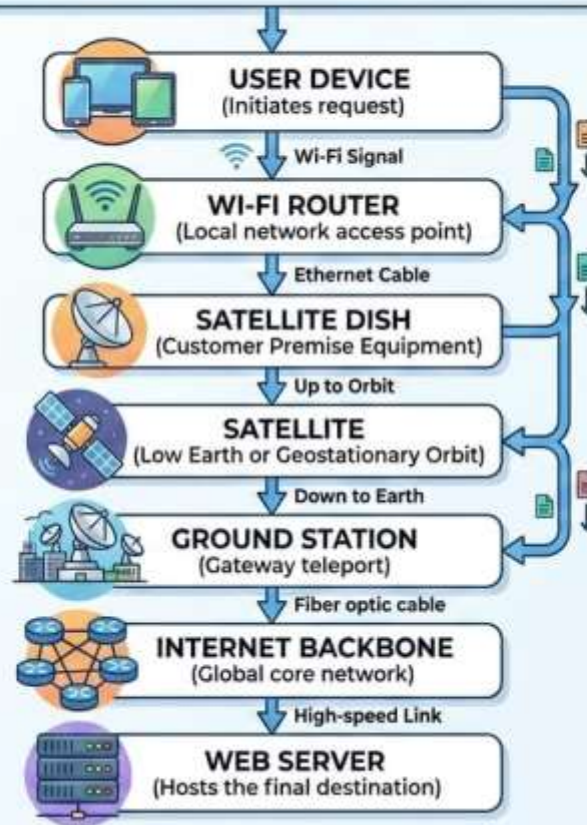


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Basic Architecture

DATA FLOW DIAGRAM: USER DEVICE TO WEB SERVER





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OSI Model Overview

Layer	Function	Layer Number (Top to Bottom)
 Application	User services	7
 Presentation	Data encryption & formatting	6
 Session	Session management	5
 Transport	Reliable communication	4
 Network	Routing	3
 Data Link	Framing & error detection	2
 Physical	Signal transmission	1



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Physical Layer



Role in Satellite Internet

- Converts data into radio waves.
- Transmits signals between Earth station and satellite.
- Uses microwave frequencies (Ku, Ka, C Band).
- Handles modulation and transmission.

Devices:

- Satellite dish
- Antenna
- Modem



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Data Link & Network



Data Link Layer:

- Creates frames
- Detects transmission errors
- Performs MAC addressing
- Controls access to the communication channel

Network Layer:

- Assigns IP addresses
- Determines routing path
- Transfers packets through satellites
- Handles fragmentation



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Transport & Session

Transport Layer:

- TCP/UDP communication
- Error recovery
- Flow control
- Retransmission of lost packets

Session Layer:

- Establishes communication session
- Synchronization
- Session recovery
- Maintains active connections



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Presentation & Application



Presentation Layer:

- Data encryption
- Compression
- Data formatting

Application Layer:

Services used by users:

- Web Browsing
- Video Streaming
- Email
- Cloud Computing
- Video Conferencing



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Advantages & Challenges



Advantages:

- Global coverage
- Works in remote locations
- High-speed internet
- Disaster recovery communication

Challenges:

- High latency (especially GEO satellites)
- Weather interference
- Expensive equipment
- Signal attenuation



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Conclusion



- Satellite Internet enables worldwide connectivity.
- Every OSI layer has a specific responsibility.
- The OSI model ensures reliable, secure, and efficient communication.
- Modern LEO satellites provide faster and lower-latency internet than traditional GEO satellites.